Drug-discovery blueprint: from scratch, with an open recipe as the reference (2026-11-05)

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.. para kahit from scratch may blueprint", same style as LIKHA). Supersedes the scope of `docs/MOLEK-V2-30M-PLAN-2026-11-05.md` *(that file stays as the Molek-only note).* Status: BLUEPRINT ONLY. Nothing trained, nothing downloaded, no spend. Every number below marked "read" was read from the open repo's own config file today; everything marked TBD is not yet measured; nothing here is a result.

1 What "from scratch" means here (honest framing, decision of 2026-10-25)

We train OUR weights from scratch on open foundations whose architecture and recipe are public, so a reference always exists to check correctness. We do not claim to have invented the architecture. Each stage first reproduces the open recipe (control), then trains ours.

2 The loop and which part is ours

target protein (FLORA / PH163 side, exists) -> pocket + affinity (Boltz-2, open, LEVERAGED) -> molecule generator (OURS: Molek v2; open DiffSBDD / REINVENT 4 as baselines) -> ADMET / toxicity (ADMET-AI / Chemprop, LEVERAGED) -> safety screen -> internal report. Only the generator is trained from scratch. Boltz-2, ADMET-AI, Chemprop are used as scorers after an REA each.

3 Reference recipes (read from the repos on 2026-11-05)

Recipe	Licence (README)	What it is	Settings read from its config
DiGress (cvignac/DiGress)	MIT	discrete graph diffusion, 2D molecules	default: graph_tf, 5 layers, dx 256 / de 64 / dy 64, 8 heads, 500 diffusion steps, cosine schedule, marginal transition, AdamW lr 2e-4, batch 512, 1000 epochs. guacamol experiment: 12 layers, dy 128, Adam lr 2e-4, batch 32, 1000 epochs. Known

Recipe	Licence (README)	What it is	Settings read from its config
			bug noted in README: `dim_ffy` ignored (default 2048 used). Original checkpoints: link lost per the README.
DeFoG (manuelml-madeira/DeFoG)	MIT	discrete flow matching for graphs, same codebase lineage as DiGress	moses experiment: graph_tf 12 layers, hidden X 256 / E 128 / y 256, 8 heads, 20 RRWP steps, AdamW lr 2e-4, batch 256, 300 epochs, remove_h true, polydec time distortion, 25,000 samples at final eval. Checkpoints provided on a shared drive.
DiffSBDD (arneschneuing/DiffSBDD)	MIT	pocket-conditioned 3D diffusion (the real structure-based drug-design recipe)	crossdock full-atom conditional: EGNN 6 layers, hidden 256, joint_nf 128, 500 steps, polynomial_2 schedule, lr 1e-3, batch 16, 1000 epochs, **4 GPUs**. 8 pretrained checkpoints on Zenodo 8183747.
MOSES (molecularsets/moses)	MIT	benchmark: data + metrics + baselines	dataset = ZINC clean leads, 1,936,962 molecules, about 1.6M train / 176k test / 176k scaffold-test; metrics valid, unique@1k/10k, novelty, FCD, SNN, Frag, Scaf, IntDiv, filters, W1 on logP/SA/QED/MW.

Not read: the papers' reported result tables (need the full text, not the README), the repos' LICENSE files themselves, parameter counts. All three must be done before any config is frozen (REA + read-recipe rules).

4 Stages

- **M0 reproduce (control).** Run DeFoG's moses config and DiGress's moses config unchanged on one RTX 4090 SECURE pod; compare our MOSES metrics to the paper's table. Pass = within the published range.

Also score DiffSBDD's published checkpoints later as the structure-based baseline. Cost to be measured on ONE epoch first.

- **M1 Molek v2-base (from scratch, 2D).** Frozen config = the reproduced recipe at the size that gives ~30M parameters (count by instantiating the model; TBD, the read configs may be larger than 30M). Data = MOSES plus licence-clean additions, deduplicated against the MOSES test and scaffold-test sets. Evaluate on MOSES test with the metrics above, pre-registered by research.
- **M2 pocket-conditioned (the actual drug-discovery generator).** DiffSBDD-class on CrossDocked and/or Binding MOAD, trained from scratch only after M1 works; the read config uses 4 GPUs for 1000 epochs, so the budget must be measured on one part first and may not fit the USD 300/week cap in one piece. Evaluate: PoseBusters-style validity, Boltz-2 rescoring (affinity), vina-type docking, vs the DiffSBDD checkpoints.
- **M3 agri/health domain targets.** Crop-protection and health targets chosen with research; ADMET-AI / Chemprop as property heads. Domain data added only after screening.

5 Datasets (licence and screen status; nothing downloaded)

MOSES (MIT per repo; ZINC-derived, check ZINC terms), GuacaMol (ChEMBL-derived; licence TBD), ChEMBL (licence TBD, believed CC BY-SA, NOT verified), CrossDocked (via the Pocket2Mol download; licence TBD), Binding MOAD (bindingmoad.org; licence TBD). For each: licence read from the source, record counts, quality screen, sufficiency and dedup report, sha256 receipt (the LIKHA pattern). No padding.

6 Tracking, in the LIKHA style

Frozen config file with sha; data receipt with sha; manifest; pre-registered metrics (research); PROGRESS lines every N steps; per-validation metrics (validity, uniqueness, novelty, loss); budget guard on the pod (MAX_USD) plus the runner wall; results shipped back and sha-verified; a technical report on every finished stage.

7 Safety floor (stated once)

A molecule generator is a dual-use tool: Urbina et al. 2022 flipped a toxicity penalty and obtained about 40,000 candidates (including VX) in six hours. Floor: known chemical-warfare agents and close analogues removed from training data, every output screened against the same list, toxicity heads never used as a maximised reward, models internal-only behind the SciACA gated tier (password, consent, audit), no ungated endpoint. This is a floor, not a full safeguard.

8 Gates before any spend

REA of each adopted repo and of Boltz-2 / ADMET-AI / Chemprop (licence files read); recipe read in full (code, not README); research pre-registration and screening pins; Demeter code review before merge; Stan's go for each paid run; one pod at a time (LIKHA holds the slot until its stage 1 ends).